

# Area Based Beaconless Reliable Broadcasting in Sensor Networks

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**Abstract.** We consider the broadcasting problem in sensor networks where the nodes have no prior knowledge of their neighborhood. That is, to preserve power and bandwidth, no beacons or 'hello' messages are sent. We describe several Area based Beaconless Broadcasting Algorithms (ABBAs). In 2D, upon receiving the packet, each node calculates the ratio  $P$  of its perimeter, along the circle of transmission radius, that is not covered by this and previous transmissions of the same packet. The node then sets or updates its timeout to be inversely proportional to  $P$ . We also consider an alternative random timeout function. If the perimeter becomes fully covered, the node cancels retransmissions. The protocol is reliable, assuming an ideal MAC layer. We also describe three 3D ABBAs, one of them being reliable, each with two choices of timeouts. These three protocols are based on covering three projections, covering particular points on intersection circles, and covering intersection points of three spheres. Our protocols are the first reliable broadcasting protocols, other than blind flooding. We compare 2D ABBAs with two other existing beaconless protocols, BPS and Geoflood, showing its superiority. We also consider a MAC layer with collisions and show that all of our methods still remain very robust, showing high delivery ratio during the broadcast.

## 1 Introduction

In this article we consider sensor networks with nodes that have the same and fixed transmission radius. Broadcasting is an important task used for paging, alarming, location updates, route discoveries or even routing in highly mobile environments. Despite of its broad use and advantages, an increase in the density produces contentions, collisions and redundancies that limit the scalability of blind flooding. There exist several localized broadcasting algorithms that use intelligent techniques to reduce communication overhead during the broadcast. However, most of these protocols assume 1-hop or 2-hop knowledge. In order to discover their neighbors, the nodes in the network use beacons. The use of

such 'hello' messages increases with mobility and changes in nodes activity status, resulting in critical bandwidth and energy losses. To address this issue, beaconless broadcasting, more intelligent than flooding, has been considered. In these protocols, the nodes act based on information gathered only during the broadcasting process. Beaconless broadcasting reduces energy consumption and increases broadcasting accuracy, since the protocols based on neighbor knowledge may have outdated information in dynamic ad hoc and sensor networks. This article studies the design of beaconless broadcasting protocols. The protocols proposed here require position information of each node to be available, which is not necessary to run blind flooding.

The first beaconless broadcasting protocols (other than blind flooding) are probabilistic and location based schemes by Ni, Tseng, Chen and Sheu [9]. However, they do not have good performance, either leaving many nodes uncovered or requiring most nodes to retransmit, as shown in [11]. We identified two existing recent beaconless broadcasting protocols: Optimized Broadcast Protocol (BPS) by Duresi, Paruchuri, Iyengar and Kannan [3] and Geoflood by Arango, Degermark, Efrat and Pink [1]. The authors claimed that their algorithm minimizes the number of necessary transmissions and that it outperforms all the other variations of flooding. However, BPS and Geoflood lack reliability.

The principal objective of this work is to develop several beaconless broadcasting algorithms. We propose four ABBAs, one for 2D and three for 3D spaces. Upon receiving the first copy of a message, each node sets a timer. Before the timeout expires, the node may receive more copies of the same message. Every node has a transmission coverage, which is assumed to be a circle in 2D and a sphere in 3D. If a node A receives a message from different sources and these sources cover the transmission circle or the sphere of A, then the node A has no need to retransmit the message. This means that each of its possible neighbors has received the message already from one or more nodes that cover it. If the node is covered before the timeout expires, then it decides not to transmit and ends the timer. We simulated our proposed broadcast algorithms, measuring the number of transmissions done for different network densities. We implemented OFP, three versions of Geoflood and compared them with 2D-ABBA. We showed that OFP and Geoflood are not reliable algorithms. In order to become reliable, Geoflood have to divide its transmission area in six angular regions instead of four. In our tests OFP failed considerably. In connected networks with average degree 7, OFP failed to complete a broadcast in 97% of the tested networks. Geoflood, even with 4 angles division, had higher retransmission counts than ABBAs. 2D-ABBA with coverage based timeout used on average 40% less messages than the reliable version of Geoflood, while the random 2D-ABBA had on average 28% less messages. Compared to Geoflood with 4 and 5 angles division, the original 2D-ABBA had on average 12% and 34% less messages respectively. Among our 3D proposals, 3D-ABBA1, based on covering three 2D projections, had the best results in terms of number of messages to complete the broadcast. However, this version lacked reliability. The second best 3D version (3D-ABBA2) performed very close to 3D-ABBA1 and had 100% delivery rate during the tests,

although this version in theory is also unreliable. 3D-ABBA3 is a reliable broadcasting protocol. It is based on a covering theorem. A sphere  $A$  is covered by other spheres if every intersection point of three spheres from the set, one of them being  $A$ , is located inside another sphere from the covering set. Any three spheres determine two such common intersection points to be covered.

Protocol versions that used random timeout function were very competitive to the original optimized timeout values. Therefore we believe that our protocols will preserve good performance if implemented under a realistic MAC layer such as in IEEE 802.11 based networks. Recall that IEEE 802.11 use random timeouts in its backoff scheme.

The remainder of this work is organized as follows. Section 2 presents the related work for the broadcasting problem. In section 3 we describe 2D-ABBA. Section 4 describes 3D-ABBAs. Section 5 compares OFP and Geoflood with 2D-ABBA. Section 6 evaluates all 3D-ABBAs. Finally section 7 concludes this article. 2D ABBA has been originally proposed in summer 2003 by last three authors, and its one paragraph description was published in [10, 6, 8]. The first three authors developed generalizations to 3D ABBAs in spring of 2004. This article summarizes master thesis work of the first author, defended in summer 2005.

## 2 Literature Review

Ni, Tseng, Chen and Sheu [9] studied the broadcast storm problem. Several beaconless schemes (probabilistic, counter-based, distance-based and location-based,) were proposed in [9] to reduce redundant rebroadcasts and differentiate the timing of rebroadcasts to alleviate this problem. In the probabilistic scheme [9], each node rebroadcasts the first copy of a received message with a given probability  $p$ . In the counter-based scheme [9], each node rebroadcasts the message if and only if it received the message from less than  $C$  neighbors. In the distance-based scheme [9], the message is retransmitted if and only if the distance to each neighbor that already retransmitted the message is greater than  $D$ . In the location-based scheme [9], the message is retransmitted if and only if the additional area that can be covered if the node rebroadcasts the message (divided by the area of circle with transmission radius) is greater than the threshold  $A$ . A simplified version of the method is to rebroadcast the message if the node is not located inside the convex hull of neighboring nodes that already retransmitted the message. However, these methods are not reliable. Further, the experimental data [9, 11] indicate low saved rebroadcasts for a high reachability, compared to other methods. We therefore did not include them in our experiments.

Bergonovo et al [2] described applications of broadcasting in inter-vehicle communications on the highway, with large and variable number of mobile terminals. The protocol appears to be a variant of the neighbor elimination scheme [11], which requires the maintenance of the neighbors list. This may not be an easy task in a highly mobile environment, the solution should be beaconless, and with guaranteed delivery.

Duresi, Paruchuri Iyengar, and Kannan [3] proposed a beaconless flooding algorithm named Optimized Broadcast Protocol (BPS). It is based on a hexagonal tiling of a plane, with the transmission radius as the edge length of the hexagons. Each broadcast packet contains two location fields L1 and L2, in its header. Whenever a node transmits a packet, it sets L1 to the location of the node from which it received the packet and sets L2 to its own location. The source node S sets both L1 and L2 to its location and transmits the packet. Each receiving node B first determines if the packet can be discarded. A packet can be discarded if the node has transmitted the same packet or if a node which is very close has already transmitted this packet. If the packet is not discarded, B determines if the packet came from the source S. If the message came from the source node S, B finds the nearest vertex V of a hexagon with center coordinates at S and with  $(S_x + R, S_y)$  as one of its vertexes. It computes its distance  $l$  from V and then delays the rebroadcast by a delay  $l/R$ . If the message came from other node K, then B selects the nearest hexagon vertex, and the rebroadcast is delay by  $1/20 \cdot R$ . After the delay elapses, B determines if the packet can be discarded. If the packet cannot be discarded, B sets L1 to the location of the node from which it received the message and L2 to its own location, and transmits. The authors of BPS propose avoiding retransmissions by having node B keep track of its distance  $dB$  to the nearest node that has retransmitted the packet. If this distance is greater than a threshold value  $Th$ , then B will retransmit. BPS authors proposed a value  $Th = 0.4R$  to ensure high delivery ratio while keeping number of transmissions low. As observed by Kim and Maxemchuk [7], the protocol may repeat flooding if the neighbors do not agree on the choice of the node near to the common ideal point. Authors [7] introduced a stopping rule to prevent that, and applied this type of flooding for route discovery. A more serious problem with BPS is its lack of reliability. As we will show in our experimental results, OFP can fail even in highly dense connected networks.

Arango, Degermark, Efrat, and Pink [1] proposed Geoflood, another beaconless broadcasting protocol. The algorithm is based on setting timers before retransmitting the message. Each node A defines a Cartesian plane with its location as the origin. Upon receiving the first copy of a message, node A records the quadrant from where the message was sent and sets a timeout. Authors [1] proposed that receiving nodes furthest away from the local sender should select smallest holding times. So timeouts increase as the distance to the sender decreases. They also introduced a random component to avoid contention that could arise between nodes located at the same distance from the sender. If node A receives more copies of the same message, at least one from each of other quadrants, before the timeout expires, it does not retransmit. The authors [1] recognized that their algorithm could fail in some situations. They showed the worst-case scenario in which part of the transmission area was not covered even when a node received messages from all four quadrants. They claimed that this area was relatively small and that other nodes in high-density networks would likely cover it. However, our experiments do not support such optimism.

Geoflood divides the transmission area in four quadrants or in four angles. This division causes the unreliability of Geoflood. In order to ensure reliability, Geoflood needs to divide into six angles instead of four. However, six transmissions from neighbors may not be necessary to cancel a retransmission, even in some cases three would be enough. This will bring unnecessary transmissions leading to unnecessary use of energy.

Recently Heissenbuttel, Braun, Wachli, and Bernoulli [4, 5] proposed a beaconless broadcasting protocol called: Dynamic Delayed Broadcasting (DDS). The idea behind DDS is basically the same to the one behind 2D-ABBA for nodes in two dimensions with position information available. Instead of taking into account perimeters as 2D-ABBA does, DDS takes into account the area left to be covered when a transmission arrives and decide not to transmit if the area left to be covered is less than a threshold value  $th$ . Unfortunately the use of  $th$ , does not guarantee delivery of the messages. The authors only proposed a 2D beaconless protocol, while we show one 2D and three 3D more sophisticated beaconless protocols.

### 3 Area Based Beaconless Algorithm in 2D

#### 3.1 Assumptions

We assume that every node in the network knows its geographic position. Our second assumption is that all the nodes have the same transmission radius and use omni-directional antennas. Finally, nodes do not know where the neighboring nodes are, so no hello messages are used.

#### 3.2 Description of 2D-ABBA

Upon receiving the first copy of a message that is broadcast, each node sets a timer. Before the timer expires, the node may receive more copies of the same message. If a node  $A$  receives a message from different sources and these sources cover the transmission circle of  $A$ , then node  $A$  has no need to retransmit the message. This means that each of its possible neighbors has received the message already from one or more nodes that cover it. If the node is covered before the timeout expires, then the node decides not to transmit and ends the timer. However, as long as the node is uncovered, the timer will continue to run, and when the timer expires the node transmits. Since all circles are of the same radius  $R$ , and each node  $A$  is aware only of the intersections with circles whose center  $B$  is inside its own circle (that is,  $|AB| \leq R$ ), the coverage criterion can be simplified. Instead of covering the whole circle centered at  $A$ , only the perimeter of that circle needs to be covered. Further, when this perimeter is intersected by several circles whose centers are inside  $A$ 's circle, there can be up to two segments on the perimeter that are not yet covered. This property further simplifies the implementation. We propose two possible timeout functions for 2D-ABBA. The first one is a function that increases when the total length of uncovered portions

of the circles with transmission radius around the node decreases. The function chosen was  $\text{timeout} = \text{degreesCovered}$ . The variable *degreesCovered* refers to the angle in degrees of the circle that has been covered. The second function that we considered is a random function with values between 0 and 1. This broadcasting protocol is reliable independently on the selected timeout function.

## 4 Area Based Beaconless Broadcasting Algorithms in 3D

### 4.1 Assumptions

We consider the transmissions patterns as spheres. All spheres are of the same radius, and each node  $A$  is only aware of transmissions from neighbors at distance  $\leq R$ . Instead of covering the whole sphere centered at  $A$ , one can consider only covering its 3D perimeter (set of nodes at distance  $= R$  from  $A$ ). All versions described here refer to covering that 3D perimeter.

### 4.2 3D-ABBA1

The idea behind our 3D-ABBA1 protocol is to use the three projections planes  $XY$ ,  $XZ$ , and  $YZ$ . By projecting the transmission spheres into the three planes, the nodes are able to apply 2D-ABBA in each plane. Node  $A$  will not transmit if its corresponding transmission circles in all three planes have been covered. This in turn is simplified to covering the corresponding 2D perimeters. Again we define two different timeout functions. The first one is directly proportional to the three perimeters (one for each plane) covered by other nodes. In other words we considered  $\text{timeout} = \frac{\text{degCovXY} + \text{degCovXZ} + \text{degCovYZ}}{3}$ . The variables *degCovXY*, *degCovXZ*, *degCovYZ* refer to the angles in degrees of the circles that have been covered in the three corresponding planes. The second function considered is a random function with values between 0 and 1.

### 4.3 3D-ABBA2

When a node  $A$  receives a transmission, a circumference of intersection is created. If after receiving several transmissions each of the created circumferences is covered by other transmission spheres then the transmission sphere of node  $A$  is completely covered by others and it will not transmit. We further simplify computationally this scheme, in two ways, leading to two protocols 3D-ABBA2 and 3D-ABBA3. They both consider certain points for coverage instead of the mentioned whole circumference. In 3D-ABBA2 variant, we propose to select six points creating a hexagon that resides in the intersection circle (circumference). Thus instead of trying to cover the whole circumference, we propose to cover six points that reside on it. We apply the following criterion at each node  $A$ . If, for every such intersection with other transmitting sphere  $B$ , each of six considered points  $P1$ -  $P6$  is located inside another transmitting sphere (different from  $A$  and  $B$ ), then the considered sphere  $A$  is covered in full. More precisely, let

$(d, e, f)$  be the coordinates of an intersection point of spheres centered at  $A$  and  $B$ . There must exist another transmitting node  $C$  with center coordinates  $(xi, yi, zi)$  such that  $(d - xi)^2 + (e - yi)^2 + (f - zi)^2 < R^2$ . Note that different selected points may be covered by different transmitting nodes. Therefore a receiving node keeps a list of all the selected points. Whenever a new transmission is received the following actions are performed. A) Some existing selected points are inside the new transmission sphere. These points are eliminated. If no point remains, the sphere is completely covered. B) The new sphere is considered, precisely the new 6 selected points. Each such point is tested whether it is inside another existing sphere (spheres of previous transmitters). Those that are inside another one are ignored. Those that are outside any existing sphere are entered into the list of points to be covered.

Estimating intersecting volumes might lead to a more precise timeout function, but the calculation is too complex to be really used. A timeout that only depends on the number of uncovered selected points will lead to simultaneous retransmissions, because of few discrete possible values for such counter. 3D-ABBA2 uses an additional parameter directly related to the volume covered by the transmission. That volume depends directly on the distance between the sender and the receiver. For this reason, we decided that each selected point has a weight  $\frac{R}{d}$ . The parameter  $d$  is the distance between the transmitter that created the point and the receiver, while  $R$  is the transmission radius. In conclusion, we propose a timeout function:  $Timeout = (\text{number of received transmissions}) + (\text{sum of the weights of the selected points still to be covered})$ .

In all our experiments, 3D-ABBA2 was always reliable. However, this is not a proof that it will always be the case. Unfortunately, there exist pathological cases where 3D-ABBA2 may fail to deliver to all the nodes connected to the source.

#### 4.4 3D-ABBA3

To design protocol 3D-ABBA3, we considered the intersections of three spheres (more precisely, their 3D perimeters) rather than two. One of these spheres is centered by node  $A$  making the retransmission decision. Thus the intersection points are on its 3D perimeter. The other two spheres are centered at two neighboring nodes  $B$  and  $C$  that transmitted the message already. The intersection (if it exists) consists of two points. If each such intersection point is located inside another transmitting sphere  $D$ , then the considered sphere is covered in full. Therefore each receiving node  $A$  keeps a list of all such uncovered intersection points. Whenever a new node  $D$  transmits a message in the neighborhood then the following events occur. A) Some existing intersection points are inside the sphere centered at  $D$ . These points are eliminated. B) The new sphere centered at  $D$  is considered, with the current fixed sphere centered at  $A$ , for intersections with any other transmitting sphere, to find new intersection points. Each such point is tested whether it is inside another transmitting sphere. Those that are inside are ignored. Those that are outside any existing sphere are entered into the list of intersection points to be covered. When a transmission from a node  $D$

does not generate new intersection points then the receiving node  $A$  adds node  $D$  to a list of transmitters that do not generate intersection points. We named this list as list of 'singular' spheres. Every time a message is received, node  $A$  checks its list of singular spheres to see if these nodes now create intersection points with the last received transmission. When an intersection is created, the corresponding node is removed from the list of singular spheres and the generated points are inserted in the list of intersection points. If the timeout for node  $A$  expires and both lists (the list of intersection points and the list of singular spheres) are empty, then node  $A$  is fully covered, and decides not to transmit. We again propose two possible timeout functions. The first one involves using the intersections created by the transmitting nodes with the receiving nodes (transmission patterns). The second is a random timeout function with values between 0 and 1. The reliability of 3D-ABBA3 follows from the following theorem.

**Theorem 1.** *Suppose that node  $A$  received transmissions from nodes  $C_1, C_2, \dots, C_m$ . Consider all the intersection points  $X$  of the 3D perimeters of the spheres centered at the three nodes:  $A, C_i$  and  $C_j$ . If there exists at least one such intersection point and every such intersection point  $X$  is located (strictly) inside at least of one of the remaining spheres, centered at  $C_k$  then the sphere centered at  $A$  is fully covered by the spheres centered at  $C_1, \dots, C_m$ , and node  $A$  does not need to retransmit, without affecting the reliability of broadcasting.*

*Proof.* Note that all the considered intersection points and all the intersection circumferences (between the sphere centered at  $A$  and any of the spheres centered at  $C_i$  for any  $i$ ) are located on the 3D perimeter of the sphere centered at  $A$ . The conditions of the theorem are then topologically equivalent to an analogous theorem for the plane, applied, for example, in [12] for sensor area coverage problems. Both theorems use circles as closed curves that are intersected, but the circles are located in the plane and on a 3D perimeter (sphere), respectively. The proof, in its simplified form, is by contradiction. Assume that the condition of the theorem is satisfied, but there is an area (on the considered 3D perimeter) that is still not covered. Let  $Y$  be a point in this area. Find the closest curve (circle) to  $Y$  from the set. This curve separates the covered and the uncovered regions. Follow this curve until another curve is met. The meeting point is an intersection point from the theorem. It is easy to see that this intersection point is not strictly inside other curve (circle), which is a contradiction.

## 5 Comparison of 2D-ABBA with BPS and Geoflood

We implemented BPS with a threshold value  $Th = 0.4R$ . For this comparison with 2D-ABBA we implemented three versions of Geoflood; one dividing the transmission area in four angles, other in five angles, and finally one with division into six angles (the reliable version of Geoflood). For the tests, we considered a network of  $n = 500$  static nodes, randomly distributed over an area of  $100 \times 100$ . We generated a total of 100 connected graphs for each of the following network degrees:  $d = 7, 8, 9, 10, 15, 20, 25, 30, 35, 40, 45, 50, 60, 70, 80, 90$ ,



100, 125. After creating each of the connected graphs we started a transmission from a randomly selected node, and we measured the following characteristics to broadcast a single message to the entire network using the original and the random 2D-ABBA, the beaconless version of OFP and the three mentioned versions of Geoflood: a) average number of transmissions, and b) average percentage of nodes receiving the message during the broadcast. We define internal node as the node whose transmitting area or volume is covered by its neighbor nodes. In other words, if before performing a broadcast the transmission pattern of a node  $A$  is covered by the transmission patterns of its neighbors, then node  $A$  may or may not retransmit during a broadcast; however, if the transmission pattern of a node  $B$  is not covered by the transmission patterns of its neighbors then node  $B$  must retransmit during a broadcast. Such type of nodes will be called as external nodes. The average percentage difference between the two versions of 2D-ABBA was 4.55%, making us believe that the random version is a viable option to be deployed in IEEE 802.11 networks. The difference between the original 2D-ABBA and BPS was 44.24%. Here besides BPS performing worst, it also suppresses transmissions from external nodes, reaffirming its unreliability. The difference between 2D-ABBA and Geoflood of 4, 5, and 6 angles was 6.83%, 20.74%, and 29.22% respectively.

For the transmitting external nodes the percentage was 100% for 2D-ABBA and for Geoflood of five and six angles. BPS causes that a considerable percentage of external nodes will not transmit, which can cause broadcast failures. In average BPS caused that 13.29% of the external nodes did not transmit during the broadcast. Even for high-connected networks (degree = 125) 30.41% of the external nodes did not transmit. These results support, that BPS can fail even in high-connected networks. In the case of Geoflood of four angles the percentage of external transmitting also was considerable. In average, 10.56% of external nodes did not transmit.

## 5.1 Impact of a Realistic MAC Layer

In order to quantify the benefits of our new protocols, we repeated our experiments on a more realistic MAC layer, which considers message collisions. We have added a contention window of size  $CW$  and a timeout for each node before it can send any message. Any node randomly picks up an integer value between 0 and  $CW$ . Then, a node can neither receive two messages at the same time nor receive a message while transmitting. We selected a contention window of size 32, in accordance to the IEEE 802.11 standard. 2D-ABBA performed better than Geoflood and BPS. This clearly shows great potential of 2D-ABBA to be deployed in IEEE 802.11 networks. Although BPS was the most unreliable algorithm, it presented less percentage of collisions. For low degree networks, the collisions contributed to the broadcast failure in all the algorithms. From degrees between 7 and 10, all algorithms failed in several occasions; however for higher degrees only BPS failed at least in one occasion.

## 6 Performance Evaluation of the 3D Area Based Beaconless Broadcasting Algorithms

We considered a network of  $n = 500$  static nodes, randomly distributed over a volume of  $100 \times 100 \times 100$ . In order to control the average node degree  $d$ , we sorted all  $\frac{n(n-1)}{2}$  (potential) edges in the network by their length, in increasing order. The transmission radius  $R$  is equal to the  $nd/2$ -th edge in the sorted array. We generated a total of 100 connected graphs for each of the following network degrees:  $d = 7, 8, 9, 10, 15, 20, 25, 30, 35, 40, 45, 50, 60, 70, 80, 90, 100$ . We considered ideal MAC and physical layers.

After creating each of the connected graphs we started a transmission from a randomly selected node, and we measured the following characteristics to broadcast a single message to the entire network using 3D-ABBA1, 3D-ABBA2, and 3D-ABBA3: a) average number of transmissions, and b) average percentage of nodes receiving the message during the broadcast. The percentage of nodes receiving the message during the broadcast was 100% in every test for all ABBA's, except for 3D-ABBA1 where we found 2 unsuccessful tests, and the average number of nodes that did not get the message during these failures was only one. This shows that although not all versions of ABBA are 100% reliable (as shown in previous sections), all versions performed extremely well on the reliability criterion. Among the 3D versions, 3D-ABBA1 performed the best in terms of number of messages needed to complete a broadcast. However, since its reliability is not 100% we cannot immediately conclude that 3D-ABBA1 is the best protocol. The second best version was 3D-ABBA2; however, this is again, in theory, an unreliable protocol. The difference between 3D-ABBA1 and 3D-ABBA3 was on average about 25% of the messages for the original timeout function and about 20% for the random timeouts. Between 3D-ABBA3 and 3D-ABBA2 the difference was on average about 3% of the messages for the original timeout function and for the random timeout versions the difference was about 5% of the messages. The original timeout function in 3D-ABBA1 had about 10% fewer re-transmissions than the random timeout function in 3D-ABBA1. However, when we applied random timeouts to 3D-ABBA2, we surprisingly obtained a minor improvement. In case of 3D-ABBA3, the difference was about 2% in favor of the original timeout function. Comparing the percentage of transmissions from internal nodes, 3D-ABBA1, 3D-ABBA2, and 3D-ABBA3 had a difference of 5.68%, 0.34%, and 2.32% between its two timer functions respectively. The difference between 3D-ABBA1 and 3D-ABBA2 was 15.11% when original timers were used and of 9.77% when random timers were tested. 3D-ABBA2 and 3D-ABBA3 had differences of 1.47% and 3.45% respectively. When high degree networks were tested only 30% of the internal nodes transmitted in 3D-ABBA3. This shows that 3D-ABBA3 achieves considerable energy savings while guaranteeing the reliable completion of a broadcast. The average difference of the percentage of transmitting external nodes was 10.80% between 3D-ABBA1 and 3D-ABBA2 (original timer functions). When using random timers, the difference was of 12.88%. The average percentage of transmitting nodes (external) shows that although 3D-ABBA1 and 3D-ABBA2 are not reliable, they both guarantee that almost all

external nodes will transmit. In 3D-ABBA1 the minimum percentage of transmitting external nodes was of 77% but in 3D-ABBA2 this minimum value was of 94%.

### 6.1 Impact of a Realistic MAC Layer in the 3D-ABBA's

As in the 2D algorithms we simulated the effect of the MAC layer using discrete random timeouts between 0 and 32 (as in IEEE 802.11). The percentage of internal transmitting increased considerably; however, we still achieved energy savings while we were able to complete a successful broadcast. This shows that our 3D proposals have potential to be deployed in IEEE 802.11 networks.

## 7 Conclusions, Future Work, and Open Ideas

Area based beaconless broadcasting algorithm (ABBA) was proposed as a broadcasting protocol for sensor networks. This algorithm has fewer assumptions than almost all broadcasting algorithms. Localized broadcasting algorithms normally assume the use of 'hello' messages to provide the list of  $k$ -hop neighbors. The use of such beacons can cause large communications overhead and delivery failures due to outdated information. Our protocols, find neighbors only when they are really needed, and rely on geographic location of the nodes. We presented four beaconless broadcasting algorithms. All versions were based in setting timeouts before retransmitting; these timeouts depended on the transmission area still to be covered. We also tested the algorithms by setting their timeouts using a discrete random function, resembling the IEEE 802.11 functionality and simulating the MAC layer impact in our algorithms.

Several beaconless routing protocols were proposed in [9] but they were not competitive in comparisons made in other articles. We propose 2D-ABBA as new beaconless broadcasting algorithm in 2D, and compare it with two recent such protocols, BPS and Geoflood. We showed that BPS is not the best broadcasting protocol since it frequently lacks reliability. We showed that Geoflood protocol, in order to be reliable, has to divide the transmission area into six angles instead of four. We implemented BPS and three versions of Geoflood in order to compare them with 2D-ABBA. 2D-ABBA had less message count than all versions of Geoflood and BPS.

The 3D-ABBA versions presented in this work are the first beaconless broadcasting algorithms proposed, other than blind flooding, to work in 3D networks. Although only one 3D-ABBA version (3D-ABBA3) is fully reliable, the other two 3D versions performed almost perfectly in the experimental tests. More precisely 3D-ABBA1 failed only in 2 tests, while 3D-ABBA2 was always reliable in our experiments. 3D-ABBA1 performed better than 3D-ABBA2, and 3D-ABBA3, in terms of the number of messages to complete a broadcast. We also measured the percentage of external and internal transmitting nodes during a broadcast. We obtained that BPS suppressed a considerable percentage of the external transmitting nodes, while in fact, the percentage of the external transmitting nodes

has to be 100% to guarantee reliability. This 100% was the case for 2D-ABBA, for Geoflood of 5 and 6 angles, and for both versions of 3D-ABBA3.

We were able to obtain excellent results with discrete random timeout variants. We believe that these show a very good promise to deploy our beaconless algorithms in IEEE 802.11 networks. Finally, we note that, besides serving for broadcasting purposes, our algorithms could also be a viable option for routing purposes.

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